

Automated Group Allocation and Report Summarization Using NLP: A Review

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Abstract. Manual project management in Undergraduate engineering programs still follow traditional, manual processes when it comes to running project work, and it shows. This includes the group formation, the assignment of mentors, and the lack of suitable tools to track student progress. This may result in the group formation with different skill levels, weak mentor guidance, slow feedback, and a lack of clear monitoring. This article explores the role that smart data-based methods play in fixing these issues. This includes the use of similarity-based matching to effectively assign mentors, language analysis to automatically process documents, and secure real-time chat and authentication systems.

Keywords: Automated Project Management · Student Group Allocation · Mentor Recommendation · Natural Language Processing · Text Summarization · Recommender Systems · Cosine Similarity · Machine Learning · Secure Authentication

1 Introduction

Project-based learning is important in the undergraduate engineering curriculum, as it helps students apply theoretical knowledge in practice to solve practical problems. However, most educational institutions still rely on traditional and opinion-based processes to manage academic projects. Group formation, mentor pairing, and document evaluation are mostly subjective processes, resulting in less effective learning.

For example, most group formation methods fail to recognize the varying skill sets of students, resulting in unbalanced groups. Similarly, the traditional method of pairing mentors with students usually does not consider the expertise of the mentors, resulting in ineffective mentoring. In addition, the traditional method of document evaluation is usually a tedious task, as documents have to be reviewed one by one. also, the lack of effective communication tools also acts

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as a barrier to effective and efficient monitoring.[1,2,3,14] Fortunately, recent developments in data-driven and intelligent technologies have provided promising solutions to these problems. For example, recommendation algorithms can be used to develop an effective and objective system for pairing mentors and students. In addition, natural language-based algorithms can be used to develop a system for evaluating academic documents.[4,5] This paper presents a review of the current methods of automating academic project management and the necessity of a single integrated system that incorporates intelligent group formation, recommendation algorithms, document evaluation, and secure communication technologies.[4,3]

2 Comparative Analysis of Existing Approaches

Table 1: Comparative analysis of AI-based approaches used in academic project management.

No.	Approach / Technology	Accuracy	Scalability
1	Document Analysis using Natural Language Techniques	High	Moderate to High
2	Resource Optimization and Scheduling using Data-Driven Methods	High	Moderate to High
3	Conversational Assistants based on Large Language Models	Moderate to High	Moderate to High
4	Intelligent Recommendation Systems for Mentor–Student Mapping	High	Moderate

3 Critical Analysis

In this section, there is a critical look at the current studies on the use of intelligent and data-driven approaches to automate higher education project management. The literature has been grouped under major themes, focusing on the methods used, their contribution, and their main strengths and weaknesses. This discussion highlights the developments and challenges faced in the design of an efficient system of academic project management.[4,8]

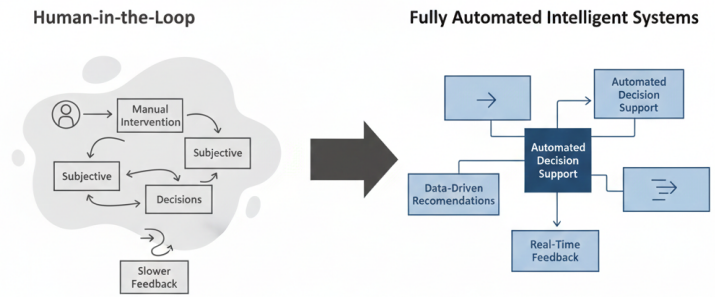


Fig. 1. Difference Between The Human Working And Automated System

3.1 Intelligent Recommendation Systems for Mentor - Student Mapping

Smart recommendation systems are essential to improving mentor-student matching by matching project requirements with what experts can provide. Unlike traditional approaches that rely on personal matching and may be influenced by personal interests and time constraints, data-driven approaches introduce objectivity and transparency to matching. Of particular interest is similarity-based approaches, especially those that utilize cosine similarity to measure how well project themes correlate with an instructor's expertise.[16,17,19]

The accuracy of these systems depends on the quality of the data. If student or faculty information is inadequate or inconsistent, then the accuracy of matching can also be affected. This is why hybrid versions of these systems, which combine automated recommendations with predefined academic constraints, can also be useful. The problem of fairness, transparency, and explainability of mentor-student matching is still a major challenge in these systems.[19,9]

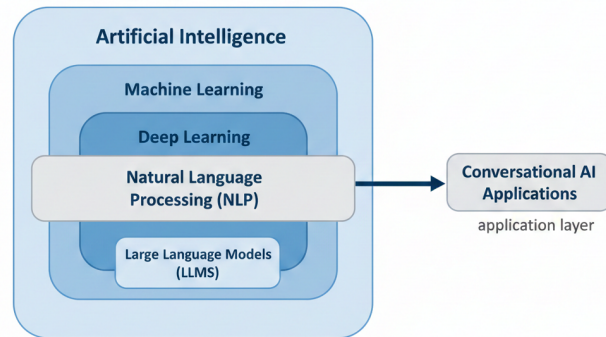


Fig. 2. The Branch Of the AI

3.2 Language-Based Project Allocation and Classification

The techniques employed in language analysis have the potential to automate text-intensive processes in academic project management, such as project allocation, evaluation of texts, and filtering of content. The techniques have demonstrated their efficiency in processing unstructured textual content such as student skill sets, project briefs, student progress reports, and scholarly texts.[6,7,10] The most advanced language models have demonstrated their potential in extracting themes, identifying key concepts, and creating concise summaries. This allows feedback to be generated automatically. that can significantly ease the workload of academics while providing uniformity in the evaluation of documents. In addition, the growth of language quality tools such as spell and grammar checkers for less common languages indicates the maturity of language techniques in supporting scholarly communication.[4,15,18]

3.3 Data-Driven Methods for Resource Optimization and Scheduling

Data-driven optimization is becoming more and more important for faculty management, project planning, and workload distribution. It goes beyond conventional rigid and static management by using predictive models and pattern recognition to make proactive decisions. For academic project management, it is used to better manage student groups, mentors, and potential delays. [2,3] Predictive models, when supported by historical data, have strong potential to predict future needs and trends of project progress. Optimization algorithms make it efficient to manage multiple constraints, such as faculty, project, and policy constraints. It is an important tool for large-scale academic institutions where it is difficult to manage everything manually. [2,3]

3.4 Security, Authentication, and Data Privacy in Academic Projects

Therefore, As universities move toward digital project management, the security of sensitive student and faculty data remains a primary concern. In this regard, the use of robust security, including token-based login and one-time password authentication, is important to locking down the system and maintaining its integrity by denying access to unauthorized entities. Cyber security studies have demonstrated the value of smart threat detection tools that can identify malicious activity, such as phishing and hacking attempts, before they have a significant impact. This is important to secure the means of communication and to secure academic records. As the trend towards online systems continues to advance, the role of security monitoring systems is becoming increasingly important.

Table 2. Point-wise summary of critical analysis

Area	Strengths	Limitations
Recommendation Systems	Objective matching; reduced bias	Data sparsity; explainability issues
Language-Based Analysis	Automated evaluation; reduced workload	sensitivity to specific subjects; loss of meaning
Optimization Methods	Predictive planning; scalability	Data dependency; low interpretability
Security Mechanisms	Strong protection	Usability trade-offs; maintenance overhead

4 Findings and Interpretation

This review shows that intelligent and data-driven methods can improve the efficiency with which academic projects are conducted, the fairness with which results are obtained, and the transparency with which processes are conducted, compared to traditional methods. Methods such as similarity-based recommendations and collaborative filtering are better at objectively assigning mentors and students by reducing bias and improving the matching between the project and the faculty.[9,19,16] Natural language analysis methods are also useful in the context of academic project management because they improve the efficiency with which academic projects are conducted by reducing the workload on the faculty and improving the consistency with which documents are assessed, depending on the diversity of the domain and the quality of the data.[4,3]

Table 3. Comparison of algorithms used in the reviewed literature

Algorithm	Key Strengths	Why Used
Cosine Similarity	Simple, interpretable	Mentor–student matching
LSTM / BiLSTM	Captures temporal patterns	Time-series prediction
NLP Models	Automates text analysis	Document evaluation
BERT-Based Models	Strong semantic understanding	Sentiment and topic analysis
OTP Authentication	Secure access control	System security

5 Changes and Limitations

5.1 Key Changes Introduced

Based on the critical review of the available literature, a number of significant changes in the approach towards academic project management systems can

be seen. First, there is a shift from a manual and availability-based student group formation and mentor allocation to a data-driven recommendation process that focuses on fairness and skill alignment. The similarity-based matching and collaborative filtering methods replace the subjective approaches to project management.[9,19] Second, there is a shift towards the automation of document-centric approaches to project management using language-based analysis methods. This shift significantly eases the burden on teachers and professors, as automated document screening, summarization, and feedback generation can be achieved.[6,15]

5.2 Limitations of Existing Approaches

In spite of these advancements, a number of limitations can be seen in the existing approaches to project management systems, based on the review of the available literature. the main limitation is that most of the approaches focus on isolated functionalities and do not support the entire project management life cycle.[3,4] The second limitation that can be seen is that most approaches to project management and project coordination rely on good and organized data to produce accurate results and to ensure the overall success of the project management process. The accuracy and effectiveness of the project management process significantly reduce in the absence of sufficient and consistent data.[6,2] The third limitation that can be seen in the available literature on project management and project coordination is that most advanced approaches to project management and project coordination lack sufficient interpretability, which limits the overall success of project management and project coordination.[6,2]

6 Conclusion

This review examines the manner in which intelligent and data-driven approaches could improve the management of academic projects in the field of higher education. The analysis indicates that recommendation systems, document processing using natural language, resource optimization algorithms, and authentication systems address the various inefficiencies associated with conventional and manually driven systems.[3,4] Nevertheless, it is also evident from the literature that the majority of the methods are not integrated at the system level, even when they are effective. This leads to the idea of the MentorLink framework, which attempts to combine student grouping, mentor allocation, document processing, real-time communication, and security into a single platform.[4,8] The literature indicates that integrated academic project management systems are essential in balancing the three important factors of automation, transparency, and usability. Such systems are an important step towards improving fairness, efficiency, and learning outcomes in undergraduate engineering education.[9,6,2]

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